Decisions in JS-Part 1

What is the ascii chart?

What is the ASCII decimal representation of the letter “A”?
65
What is the ASCII decimal representation of the letter “a”?
97
What is the ASCII decimal representation of the character “)”?
41
What is the ASCII decimal representation of the character “#”?
35
State whether each condition is true or false by writing out the condition using the english language (the first is done for you):

```
5 > 3  five is greater than three is true
10 == 10   T
"hello" != "world"   T
7 <= 2     F
false == false  T
3 >= 4     F
"" == ""   T
'a' == 'b' F
100 != 100 F
"apple" == "apple"    T
"cat" != "dog"  T
"hello" == "Hello"    F
"test" == "testing"   F
"123" != "321"  T
"" == ""   T
"space " == "space"   F
"true" == "false"     F
"abc" < "abd"   T
```

Solve the following problems. Fill in the blanks to the partially solved problems.

Exercise : Divisible by 10?

Write a program that takes in a number and checks if the number is divisible by 10.

```
let number=+prompt("Enter the  number:");
let result=number%10;
if(result==_0_____)
{
   console.log("The number is divisible by 10!");
}
else
{
   console.log("The number IS NOT divisible by 10.");
}
```

Exercise : Consecutive Number

Write a program that takes in a number and either a `+` or a `-`. The program should return the consecutive number either before (`+`) or after (`-`) the number. For example, if someone enters 3 and + then show 4.

```
let num = +prompt("Enter a number:");
let sign = prompt("Enter '+' or '-' :");
```

```
if (sign == "+") {
  console.log(num + 1);
} else if (sign == "-") {
  console.log(num - 1);
} else {
  console.log("Invalid input!");
}
```

Exercise : Sum Less than 100?

Write a program that takes in two numbers and checks if the sum of the two numbers is less than 100.

```
let number1=+prompt("Enter the  number:");
let number2=+prompt("Enter the second number:");
let _sum____=number1+number2;


if(sum<100)
{
   console.log("Their sum is less than 100.");
}
else
{
   console.log("Their sum IS NOT less than 100.");
}
```

Exercise : Is the Number Even?

Write a program that takes in an integer and checks if the number is even.

```
let n = +prompt("Enter a number:");
if (n % 2 == 0) {
  console.log("The number is even.");
} else {
  console.log("The number is odd.");
}
```

Exercise : Divisible by a Number

Write a program that takes in two integers. The program will check if the first integer is divisible by the second integer.

```
let first = +prompt("Enter the first integer:");
let second = +prompt("Enter the second integer:");

if (first % second == 0) {
  console.log(first + " is divisible by " + second);
} else {
  console.log(first + " is NOT divisible by " + second);
}
```

Exercise : Perpendicular Lines

Lines are perpendicular if their slopes are negative reciprocals. Write a program that takes in the numerators and denominators of the slope of two lines and checks if the lines are perpendicular. Given two fractions, a/b = c/d , lines are perpendicular if a/b = - d/c .

```
let a = +prompt("Enter numerator of first slope:");
let b = +prompt("Enter denominator of first slope:");
let c = +prompt("Enter numerator of second slope:");
let d = +prompt("Enter denominator of second slope:");

if (a / b == -d / c) {
  console.log("The lines are perpendicular.");
} else {
  console.log("The lines are not perpendicular.");
}
```

Exercise : At Capacity

Write a program that determines if a stadium is at maximum capacity. A stadium is at capacity if at least 80% of the seats are sold out.

```
let totalSeats = +prompt("Enter total number of seats:");
let soldSeats = +prompt("Enter number of seats sold:");

let percentSold = (soldSeats / totalSeats) * 100;
```

```
if (percentSold >= 80) {
  console.log("The stadium is at capacity!");
} else {
  console.log("The stadium is not at capacity.");
}
```

Exercise : Enough Paint

Write a program that indicates if an artist has enough paint to paint their ceiling. `The program` takes two inputs, the number of gallons of paint, and the square footage of the ceiling. 1 gallon of paint is enough to cover 5 square feet.

```
let gallons = +prompt("Enter number of paint gallons:");
let squareFeet = +prompt("Enter ceiling square footage:");

let coverage = gallons * 5;

if (coverage >= squareFeet) {
  console.log("You have enough paint!");
} else {
  console.log("You need more paint.");
}
```

Exercise : Can we Make it?

Write a program that determines whether or not a driver can make it a specific distance in miles based on their current amount of gas and the car's miles per gallon.

```
let milesPerGallon=+prompt("How many miles per gallon can your car drive?:");

let gallons=+prompt("How many gallons of gas does your car have?");

let distance=+prompt("How many miles would you like your car to travel?");

let totalPossibleMiles=_milesPerGallon_____*gallons;

if(totalPossibleMiles>_=distance_____)

{

   console.log("You will be able to make it!");

}

else

{

   console.log("Your car needs more fuel as it will not make this distance based on its fuel efficiency and fuel amount.");

}
```

Exercise : Tax Rate

Write a program that adds taxes to the cost of a purchase depending on what province you're in. The three provinces that we will use for this exercise will be Ontario(13.00% sales tax), Quebec (6.5%), and Manitoba (5.5%). If one of those states is not given as an input, then the tax rate added will be 7%.

Use the provinces abbreviation (Ontario = ON) for the input of the province.

```
let province = prompt("Enter your province (ON / QC / MB):");
let cost = +prompt("Enter purchase cost:");
let taxRate = 0;

if (province == "ON") {
  taxRate = 0.13;
} else if (province == "QC") {
  taxRate = 0.065;
} else if (province == "MB") {
  taxRate = 0.055;
} else {
  taxRate = 0.07;
}

let total = cost + (cost * taxRate);
console.log("Total cost after tax: $" + total.toFixed(2));
```

Exercise : +,- or 0?

- Using if..else, write the code which gets a number via prompt and then shows in alert:

1, if the value is greater than zero,

-1, if less than zero,

0, if equals zero.

```
let number = +prompt("Enter a number:");


if (number > 0) {
  alert(1);
} else if (number < 0) {
  alert(-1);
```

```
} else {
  alert(0);
}
```

Exercise : Max Number

Write a program to find the maximum between two numbers using if else.

Example
Input

```
Input num1: 10
Input num2: 20
```

Output

```
Maximum = 20
```

```
let num1 = +prompt("Enter first number:");
let num2 = +prompt("Enter second number:");

if (num1 > num2) {
  console.log("Maximum = " + num1);
} else {
  console.log("Maximum = " + num2);
}
```

Exercise : Your Grade?

- Write a program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer, calculate percentage and grade according to given conditions:
  If percentage >= 90% : Grade A
  If percentage >= 80% : Grade B
  If percentage >= 70% : Grade C
  If percentage >= 60% : Grade D

If percentage >= 40% : Grade E
If percentage < 40% : Grade F

Example

Input

```
Input marks of five subjects: 95 95 97 98 90
```

Output

```
Percentage = 95.00 Grade A
```

```
let physics = +prompt("Enter Physics marks:");
let chemistry = +prompt("Enter Chemistry marks:");
let biology = +prompt("Enter Biology marks:");
let math = +prompt("Enter Mathematics marks:");
let computer = +prompt("Enter Computer marks:");

let percentage = (physics + chemistry + biology + math + computer) / 5;

console.log("Percentage = " + percentage.toFixed(2) + "%");

if (percentage >= 90) console.log("Grade A");
else if (percentage >= 80) console.log("Grade B");
else if (percentage >= 70) console.log("Grade C");
else if (percentage >= 60) console.log("Grade D");
else if (percentage >= 40) console.log("Grade E");
else console.log("Grade F");
```

Exercise : Is it Equal?

- Write a program to accept two integers and check whether they are equal or not.
  Test Data : 15 15
  Expected Output :
  Number1 and Number2 are equal

```
let number1 = +prompt("Enter first integer:");
let number2 = +prompt("Enter second integer:");

if (number1 == number2) {
  console.log("Number1 and Number2 are equal");
} else {
```

```
  console.log("Number1 and Number2 are not equal");
}
```

Exercise : Odd or Even?

- Write a program to check whether a given number is even or odd.

Test Data : 15
Expected Output :
15 is an odd integer

```
let num = +prompt("Enter a number:");

if (num % 2 == 0) {
  console.log(num + " is an even integer.");
} else {
  console.log(num + " is an odd integer.");
}
```

Exercise : Voting?

- Write a program to read the age of a candidate and determine whether it is eligible for casting his/her own

Test Data : 21
Expected Output :
Congratulation! You are eligible for casting your vote.

```
let age = +prompt("Enter your age:");

if (age >= 18) {
  console.log("Congratulations! You are eligible for casting your vote.");
} else {
  console.log("Sorry, you are not eligible to vote yet.");
}
```